For the logistic regression:

LOGISTIC REGRESSION RESULTS

Accuracy: 0.6842
McFadden pseudo-R-squared: 0.0625

Classification Report:

	precision	recall	f1-score	support
0	0.69	0.96	0.81	26
1	0.50	0.08	0.14	12
accuracy			0.68	38
macro avg	0.60	0.52	0.47	38
weighted avg	0.63	0.68	0.60	38

Confusion Matrix (actual rows, predicted columns):
[[25 1]
[11 1]]

1. What is the model's accuracy? The model's accuracy is 68.42%.
2. What is the model's lowest precision? The model's lowest precision is 50% for class 1.
3. Do all categories have at least 30 support? (confusion matrix)
   No, not all categories have 30 support. Class 0 has 26 and class 1 has 12.

The confusion Matrix is [[25 1] [11 1]]

4. Based on the pseudo-R^2 value, is this model good in this scenario? A pseudo R^2 of 0.0624 so I would say it's not a great model in this scenario because only 1 out of the 12 actual low birthweight was identified.

For the count-variable regression:

POISSON REGRESSION RESULTS

Generalized Linear Model Regression Results
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Dep. Variable: mhlth_crudeprev No. Observations: 2299
Model: GLM Df Residuals: 2287

Model Family: Poisson Df Model: 11
Link Function: Log Scale: 1.0000
Method: IRLS Log-Likelihood: -5397.6
Date: Sun, 25 Oct 2026 Deviance: 39.316
Time: 23:02:03 Pearson chi2: 39.2
No. Iterations: 4 Pseudo R-squ. (CS): 0.1925
Covariance Type: nonrobust

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.7032	0.172	9.888	0.000	1.366	2.041
binge_crudeprev	0.0032	0.003	0.954	0.340	-0.003	0.010
checkup_crudeprev	-0.0004	0.002	-0.192	0.848	-0.004	0.004
csmoking_crudeprev	0.0043	0.004	1.100	0.271	-0.003	0.012
depression_crudeprev	0.0137	0.002	5.800	0.000	0.009	0.018
obesity_crudeprev	0.0001	0.002	0.076	0.940	-0.003	0.004
sleep_crudeprev	0.0024	0.002	1.059	0.289	-0.002	0.007
selfcare_crudeprev	-0.0608	0.022	-2.708	0.007	-0.105	-0.017
indeplive_crudeprev	0.0522	0.014	3.759	0.000	0.025	0.079
loneliness_crudeprev	0.0113	0.003	3.717	0.000	0.005	0.017
foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
housinsecu_crudeprev	0.0150	0.008	1.786	0.074	-0.001	0.032

Pseudo-R-squared:
0.1925

Variance Inflation Factors (VIF):
binge_crudeprev: 2.5288
checkup_crudeprev: 1.7504
csmoking_crudeprev: 6.9348
depression_crudeprev: 2.2164
obesity_crudeprev: 2.5730
sleep_crudeprev: 3.2093
selfcare_crudeprev: 30.5144
indeplive_crudeprev: 30.9533
loneliness_crudeprev: 3.4326
foodinsecu_crudeprev: 62.8308
housinsecu_crudeprev: 46.8400

Significant predictors ($p < 0.05$):
depression_crudeprev: coefficient = 0.013653, p-value = 6.6371e-09
selfcare_crudeprev: coefficient = -0.060769, p-value = 0.00676918
indeplive_crudeprev: coefficient = 0.052157, p-value = 0.000170627
loneliness_crudeprev: coefficient = 0.011268, p-value = 0.000201438

1. Your model should include 4 significant predictors. What are they?
    1. Depression_crudeprev
    2. Selfcare_crudeprev
    3. Indeplive_crudeprev
    4. Loneliness_crudeprev
2. Which of these predictors is the most significant (hint: look at the 'coef' column)
   The most significant is selfcare_crudeprev
3. Do any of your significant predictors have an overinflated VIF (VIF > 10)?
   Yes, selfcarecrudeprev *has a VIF of approximately 30.51*, and indeplive_crudeprev has a VIF of approximately 30.95.
4. How much of the variability is captured by the model (hint: look at pseudo-r^2)

About 19.25% or 0.1925.